

	$\sim 1200 \text{ N}$	
3(a)	Internal energy is <u>determined by the state of the system</u> and it can be expressed as <u>the sum of a random distribution of kinetic and potential energies associated with the molecules of the system.</u>	[1] for correct answer
3(b)	$p = \frac{1}{3} \rho \langle c^2 \rangle$ $= \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$ $\Rightarrow pV = \frac{1}{3} Nm \langle c^2 \rangle \dots\dots\dots (1)$ Subst $pV = nRT$ into (1) $\frac{1}{3} Nm \langle c^2 \rangle = nRT$ $\Rightarrow \frac{1}{2} m \langle c^2 \rangle = \frac{3}{2} \frac{n}{N} RT \dots\dots\dots (2)$ $\Rightarrow E_k \propto T$ Hence, the average kinetic energy of a molecule of an ideal gas is proportional to its thermodynamic temperature T. <i>Comments:</i> <i>A significant number of the candidates concluded on the kinetic energy of the system instead of the average kinetic energy of a molecule.</i>	[1] for showing the derivation of equation (1) [1] for equating ideal gas equation correctly and correct workings.

No.	Solution				Remarks
3(c)(i)1.	Using $\Delta U = Q + W_{on}$ and $U = \frac{3}{2}nRT$, $\frac{3}{2}nR\Delta T = Q + (0) \quad (\because B \rightarrow C \text{ constant volume})$ $\frac{3}{2}(0.024)(8.31)\Delta T = 250$ $\Delta T = 835.67 \text{ K}$ $\therefore T = 976 + 835.67 \approx 1810 \text{ K}$ <i>Comments:</i> Common mistakes include using the wrong moles; treating ΔT as the final temperature.				[1] for showing $\Delta U = Q + W_{on}$ $\frac{3}{2}nR\Delta T = Q + (0)$ [1] for correct substitution and ΔT [1] for correct T
3(c)(i)2.	Using $pV = nRT$, $p = \frac{nRT}{V}$ $= \frac{0.024 \times 8.31 \times 1811.67}{0.32 \times 10^{-4}}$ $= 11291233$ $\approx 1.13 \times 10^7 \text{ Pa}$				[1] for correct answer
3(c)(ii)	change	heat supplied to gas / J	work done by gas / J	increase in internal energy / J	[3] for all correct answers Deduct 1 mark for every 2 wrong answers
	A to B	0	-200	200	
	B to C	250	0	250	
	C to D	0	350	-350	
	D to A	-100	0	-100	
	<i>Comments:</i> Common mistake includes treating "work done by" as "work done on".				
4(c)(i)	The magnetic force provides the centripetal force required to keep				